CTI / RASD — Data Model

This document covers the star schema behind the CTI / RASD analytics layer:

1. Entity-relationship diagram: dimensions, fact and bridges
2. Model table descriptions
3. Semantic layer: the `vw_*` views built on top of the model

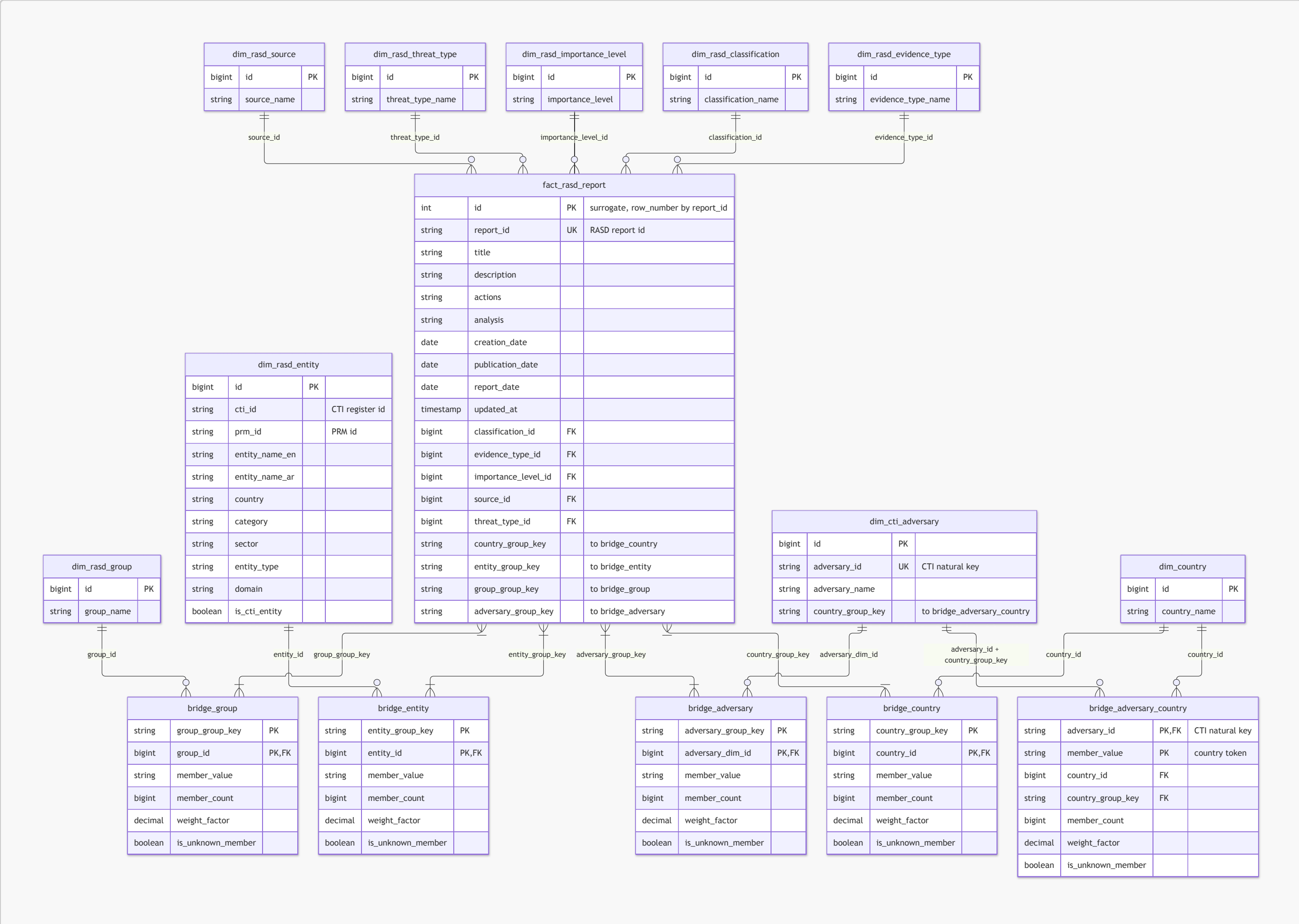
Schema	`cti_gold` (Spark / Hive metastore, Delta Lake)
Source schema	`cti` (`rasd`, `adversary`, `entities`)
Source SQL	`sql/dim`, `sql/fact`, `sql/bridge`, `sql/semantic`
Orchestration	Airflow DAG `spark_cti_rasd_data_model_dag`, daily at 03:00
Build order	staging → `dim_key` → dimensions → fact + bridges → semantic views

Modelling conventions

Convention	Meaning
Surrogate keys	Every dimension has an `id BIGINT` taken from the key registry `dim_key`. The registry is append-only: a value keeps its id from the day it is first seen, even though dimension tables are fully rebuilt each run. Ids are numbered per dimension, so the same number means different things in different dimensions.
Natural keys	Matching is case-insensitive: a dimension value is identified by `upper(label)`. The stored label is the spelling seen when the value was first registered.
Unknown member	Each dimension has an `id = -1` row labelled `'Unknown'`. The fact and the bridges never contain `NULL` foreign keys; missing or unmatched values point to `-1`.
Missing values	In the source, `'NULL'`, `'NUL'` and empty strings are treated as missing.
Group keys	A report can have many countries, entities, groups and adversaries. Each such set is stored on the fact as `*_group_key = md5(sorted upper-cased members joined by '|')`. An empty set becomes `['UNKNOWN']`. Reports with the same set share a key, and the bridge holds one row per key × member.
Weighting	Bridge `weight_factor = 1 / member_count`, so one report's weights across a bridge add up to 1.
Load strategy	Dimensions, fact and bridges are full refreshes (`INSERT OVERWRITE`); `dim_key` is appended; views are `CREATE OR REPLACE VIEW`.

1. Entity-relationship diagram

Scope: dimension, fact and bridge tables only. Staging, `dim_key` and views are left out.



Relationship notes

Relationship	Cardinality	Join	Notes
RASD dimensions → `fact_rasd_report`	1 : N	`dim.id = fact.<dim>_id`	Single-valued report attributes. Missing → `-1`.
`fact_rasd_report` ↔ `bridge_country` / `bridge_entity` / `bridge_group` / `bridge_adversary`	N : M	`fact.<x>_group_key = bridge.<x>_group_key`	A group key isn't unique on either side: many reports share a member set, and a set has many members. Every fact key has at least one bridge row (possibly the Unknown member).
`dim_country` → `bridge_country`	1 : N	`dim_country.id = bridge_country.country_id`	
`dim_rasd_entity` → `bridge_entity`	1 : N	`dim_rasd_entity.id = bridge_entity.entity_id`	
`dim_rasd_group` → `bridge_group`	1 : N	`dim_rasd_group.id = bridge_group.group_id`	
`dim_cti_adversary` → `bridge_adversary`	1 : N	`dim_cti_adversary.id = bridge_adversary.adversary_dim_id`	

Relationship	Cardinality	Join	Notes
`dim_cti_adversary` → `bridge_adversary_country`	1 : N	`adversary_id` **and** `country_group_key`	Joins on the CTI natural key, not the surrogate `id`. Adversaries with the same target set share a `country_group_key`, so `adversary_id` is required.
`dim_country` → `bridge_adversary_country`	1 : N	`dim_country.id = bridge_adversary_country.country_id`	`dim_country` is shared by RASD reports and CTI adversaries.

Typical star join through a bridge

```
SELECT c.country_name,
       COUNT(DISTINCT f.report_id) AS reports,
       SUM(b.weight_factor)        AS weighted_reports
FROM cti_gold.fact_rasd_report f
JOIN cti_gold.bridge_country   b ON b.country_group_key = f.country_group_key
JOIN cti_gold.dim_country      c ON c.id = b.country_id
GROUP BY c.country_name;
```

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2. Model table descriptions

2.1 Fact

`fact_rasd_report`

Description: One record per RASD threat report. It holds the report text, its dates, foreign keys to the five single-valued RASD dimensions, and group keys to the four multi-valued relationships. **Grain:** One row per report (`report_id`). **Source:** `cti.rasd` where `report_type = 'rasd'` (through `stg_report`). **Primary key:** `id`. **Business key:** `report_id`.

Column	Type	Key	Description
`id`	INT	PK	Surrogate key, `row_number()` ordered by `report_id`. Renumbered on every rebuild, so not stable over time.
`report_id`	STRING	UK	RASD report identifier.
`title`	STRING		Report title.
`description`	STRING		Narrative description.
`actions`	STRING		Recommended or taken actions.
`analysis`	STRING		Analyst assessment.
`creation_date`	DATE		Creation date. Accepts ISO, `dd/MM/yyyy[ HH:mm:ss]` and `MM/dd/yyyy`; `NULL` if unparseable.
`publication_date`	DATE		Publication date (same parsing).

Column	Type	Key	Description
`report_date`	DATE		Report / observation date (same parsing).
`updated_at`	TIMESTAMP		Last update in source.
`classification_id`	BIGINT	FK	→ `dim_rasd_classification.id`
`evidence_type_id`	BIGINT	FK	→ `dim_rasd_evidence_type.id`
`importance_level_id`	BIGINT	FK	→ `dim_rasd_importance_level.id`
`source_id`	BIGINT	FK	→ `dim_rasd_source.id`
`threat_type_id`	BIGINT	FK	→ `dim_rasd_threat_type.id`
`country_group_key`	STRING(32)		MD5 of the report's sorted country set → `bridge_country`. From `rasd.country` (comma-separated).
`entity_group_key`	STRING(32)		MD5 of the report's sorted entity set → `bridge_entity`. From `rasd.entities_cti_id`, or `rasd.entities_names_ar` if that is empty (pipe-separated).
`group_group_key`	STRING(32)		MD5 of the report's sorted threat-group set → `bridge_group`. From `rasd.related_groups` (pipe-separated).
`adversary_group_key`	STRING(32)		MD5 of the sorted ids of CTI adversaries whose `rasd_ids` list this report → `bridge_adversary`.

2.2 Dimensions

All dimensions include the Unknown row (`id = -1`). Unless stated otherwise they are built as `dim_key` rows for that dimension (`id`, `label`) plus that Unknown row.

`dim_rasd_classification`

Description: Report classification values. **Grain:** One row per distinct classification (case-insensitive). **Source:** `rasd.classification`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key from `dim_key`; `-1` = Unknown.
`classification_name`	STRING		Classification label.

`dim_rasd_evidence_type`

Description: Types of evidence that support a report. **Grain:** One row per distinct evidence type. **Source:** `rasd.evidence_type`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown.
`evidence_type_name`	STRING		Evidence type label.

`dim_rasd_importance_level`

Description: Importance levels given to reports. **Grain:** One row per distinct importance level. **Source:** `rasd.importance`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown. Ids follow first-seen order, **not** severity order.
`importance_level`	STRING		Importance label.

`dim_rasd_source`

Description: Where the observation came from. **Grain:** One row per distinct observation source. **Source:** `rasd.observation_source`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown.
`source_name`	STRING		Observation source label.

`dim_rasd_threat_type`

Description: Threat categories. **Grain:** One row per distinct threat type. **Source:** `rasd.threat_type`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown.
`threat_type_name`	STRING		Threat type label.

`dim_rasd_group`

Description: Threat groups named in RASD reports. **Grain:** One row per distinct group name. **Source:** `rasd.related_groups` (multi-valued). **Used by:** `bridge_group`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown.
`group_name`	STRING		Threat group name.

`dim_country`

Description: Shared country dimension, used by both RASD reports and CTI adversary targeting. **Grain:** One row per distinct country name (case-insensitive). **Source:** `rasd.country` (comma-separated) and `adversary.targeted_country` (pipe-separated). **Used by:** `bridge_country`, `bridge_adversary_country`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown.
`country_name`	STRING		Country name as first registered.

`dim_rasd_entity`

Description: Organisations referenced by reports. It combines the CTI entity register with label-only entities, which appear in report text but aren't in the register. **Grain:** One row per entity natural key: `upper(coalesce(cti_id, entity_name_ar))`. **Source:** `cti.entities` (registered, one row per `cti_id`) plus Arabic names from `rasd.entities_names_ar` for reports without `entities_cti_id` that don't match a registered Arabic name. **Used by:** `bridge_entity`.

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown.
`cti_id`	STRING		CTI entity register id; `NULL` for label-only entities.
`prm_id`	STRING		PRM system id; `NULL` if not linked.
`entity_name_en`	STRING		English name. For label-only entities this repeats the Arabic name.
`entity_name_ar`	STRING		Arabic name (`'غير معروف'` for Unknown).
`country`	STRING		The entity's country, from the register (free text, not linked to `dim_country`).
`category`	STRING		Entity category.
`sector`	STRING		Entity sector.
`entity_type`	STRING		Type of entity (`entities.type_of_entity`).
`domain`	STRING		Entity web domain.
`is_cti_entity`	BOOLEAN		`TRUE` = from the CTI register; `FALSE` = label-only or Unknown.

`dim_cti_adversary`

Description: Threat adversaries from CTI intelligence. **Grain:** One row per CTI adversary. **Source:** `cti.adversary` (through `stg_adversary`). **Used by:** `bridge_adversary` (on `id`), `bridge_adversary_country` (on `adversary_id` + `country_group_key`).

Column	Type	Key	Description
`id`	BIGINT	PK	Surrogate key; `-1` = Unknown.
`adversary_id`	STRING	UK	CTI adversary identifier (natural key); `NULL` for Unknown.
`adversary_name`	STRING		Adversary name.
`country_group_key`	STRING(32)		MD5 of the adversary's sorted, upper-cased target countries (`md5('Unknown')` if none) → `bridge_adversary_country`.

2.3 Bridges

Every bridge is built the same way. Take the distinct member sets (from `stg_report`, or `stg_adversary` for `bridge_adversary_country`), explode them into members, match each member to its dimension, and group by (group key, dimension id). Members that don't match all become **one** row with dimension id `-1`. The exception is `bridge_adversary_country`, which keeps one row per token.

Common column	Type	Description
`member_value`	STRING	Upper-cased source token that was matched (the smallest one if several matched the same id).
`member_count`	BIGINT	Number of rows (distinct matched ids) for the group key.
`weight_factor`	DECIMAL(23,22)	`1 / member_count`.
`is_unknown_member`	BOOLEAN	`TRUE` when the dimension id is `-1`.

`bridge_country`

Description: Resolves a report's country set into individual `dim_country` members. **Grain:** One row per `country_group_key` × `country_id`. **Joins:** `fact_rasd_report.country_group_key`, `dim_country.id`.

Column	Type	Key	Description
`country_group_key`	STRING(32)	PK	Report country-set key.
`country_id`	BIGINT	PK, FK	→ `dim_country.id`; `-1` = unmatched or no country.
`member_value`	STRING		Upper-cased country token from the report.
`member_count`	BIGINT		Countries in the set.
`weight_factor`	DECIMAL(23,22)		`1 / member_count`.
`is_unknown_member`	BOOLEAN		`country_id = -1`.

`bridge_entity`

Description: Resolves a report's entity set into `dim_rasd_entity` members. Each token is matched in priority order: `cti_id` → `prm_id` → `entity_name_ar` → `entity_name_en` (case-insensitive; lowest `id` wins ties). **Grain:** One row per `entity_group_key` × `entity_id`. **Joins:** `fact_rasd_report.entity_group_key`, `dim_rasd_entity.id`.

Column	Type	Key	Description
`entity_group_key`	STRING(32)	PK	Report entity-set key.
`entity_id`	BIGINT	PK, FK	→ `dim_rasd_entity.id`; `-1` = unmatched or no entity.
`member_value`	STRING		Upper-cased entity token (CTI id or Arabic name).
`member_count`	BIGINT		Entities in the set.
`weight_factor`	DECIMAL(23,22)		`1 / member_count`.
`is_unknown_member`	BOOLEAN		`entity_id = -1`.

`bridge_group`

Description: Resolves a report's threat-group set into `dim_rasd_group` members. **Grain:** One row per `group_group_key` × `group_id`. **Joins:** `fact_rasd_report.group_group_key`, `dim_rasd_group.id`.

Column	Type	Key	Description
`group_group_key`	STRING(32)	PK	Report group-set key.
`group_id`	BIGINT	PK, FK	→ `dim_rasd_group.id`; `-1` = unmatched or no group.
`member_value`	STRING		Upper-cased group token.
`member_count`	BIGINT		Groups in the set.
`weight_factor`	DECIMAL(23,22)		`1 / member_count`.
`is_unknown_member`	BOOLEAN		`group_id = -1`.

`bridge_adversary`

Description: Resolves the set of CTI adversaries linked to a report into `dim_cti_adversary` members. An adversary is linked when its `adversary.rasd_ids` list contains the report id. **Grain:** One row per `adversary_group_key` × `adversary_dim_id`. **Joins:** `fact_rasd_report.adversary_group_key`, `dim_cti_adversary.id`.

Column	Type	Key	Description
`adversary_group_key`	STRING(32)	PK	Report adversary-set key.
`adversary_dim_id`	BIGINT	PK, FK	→ `dim_cti_adversary.id`; `-1` = no linked or unmatched adversary.
`member_value`	STRING		CTI `adversary_id` token (`'UNKNOWN'` when none).
`member_count`	BIGINT		Adversaries in the set.
`weight_factor`	DECIMAL(23,22)		`1 / member_count`.
`is_unknown_member`	BOOLEAN		`adversary_dim_id = -1`.

`bridge_adversary_country`

Description: Links each CTI adversary to the countries it targets. This bridge is **not** report-based. **Grain:** One row per `adversary_id` × country token (`member_value`). Unmatched tokens are not grouped, so one adversary can have several `country_id = -1` rows. **Source:** `stg_adversary.country_members` (from `adversary.targeted_country`, pipe-separated). **Joins:** `dim_cti_adversary` on (`adversary_id`, `country_group_key`); `dim_country.id`.

Column	Type	Key	Description
`adversary_id`	STRING	PK, FK	→ `dim_cti_adversary.adversary_id` (natural key).
`member_value`	STRING	PK	Upper-cased country token.
`country_id`	BIGINT	FK	→ `dim_country.id`; `-1` = unmatched country.
`country_group_key`	STRING(32)	FK	Adversary target-set key, equal to `dim_cti_adversary.country_group_key`.
`member_count`	BIGINT		Country tokens listed for the adversary (matched and unmatched).
`weight_factor`	DECIMAL(23,22)		`1 / member_count`.
`is_unknown_member`	BOOLEAN		`country_id = -1`.

Adversaries with no target country have no rows in this bridge.

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3. Semantic layer

The semantic layer is the set of `vw_*` views that analysts, dashboards and the Django app query. The views sit on top of the star schema (`fact_rasd_report`, the `dim_*` tables and the `bridge_*` tables). They swap surrogate keys for readable labels and flatten the many-to-many links, so consumers don't need to know how the bridges work.

Conventions

These rules apply to every view below.

Convention	Meaning
`'Unknown'` labels	A report attribute that is missing in the source (classification, importance, threat type, ...) shows `'Unknown'`, never `NULL`. It maps to the dimension row with `id = -1`.
`id = -1`	Surrogate id of the *Unknown* member in every dimension.

Convention	Meaning
`is_unknown_member`	`TRUE` when the bridge member couldn't be matched to a dimension row. Several unmatched values in one report collapse into a **single** row with id `-1`.
`weight_factor`	`1 / number of distinct resolved members` in the member set. For one report, the weights across its rows add up to `1`. Use it to split a report across its countries, entities, groups or adversaries without double counting.
`*_group_key`	32-character MD5 hex of the sorted, upper-cased member list (for example, all countries on a report). Reports with the same member set share a key. It is a technical join key, not a business attribute.
Dates	`creation_date`, `publication_date` and `report_date` are parsed from ISO, `dd/MM/yyyy HH:mm:ss`, `dd/MM/yyyy` or `MM/dd/yyyy`. Values that fail to parse are `NULL`.

Counting vs. weighting

Member views return one row **per report × member**, so a report that names three countries shows up three times.

```sql
-- Reports that mention each country (a report counts once for every country it names)
SELECT country_name, COUNT(DISTINCT report_id) AS reports
FROM cti_gold.vw_report_country
GROUP BY country_name;

-- Weighted share: each report contributes a total of 1, split across its countries
SELECT country_name, SUM(weight_factor) AS weighted_reports
FROM cti_gold.vw_report_country
GROUP BY country_name;
```

Never use `COUNT(*)` on a member view to count reports.

View catalog

View	Grain (one row per ...)	Answers
`vw_report`	report	What does each report say, and how is it classified?
`vw_report_country`	report × country	Which countries does each report concern?
`vw_report_entity`	report × entity	Which organisations does each report concern?
`vw_report_group`	report × threat group	Which threat groups are named in each report?
`vw_report_adversary`	report × CTI adversary	Which CTI adversaries are linked to each report?
`vw_adversary_target_country`	adversary × target country	Which countries does each adversary target (per CTI, not per report)?
`vw_entity_coverage`	metric	How well do report entities match the CTI entity register?

```
vw_report ─┬─ vw_report_country
           ├─ vw_report_entity ── vw_entity_coverage
           ├─ vw_report_group
```

```
            └─ vw_report_adversary

vw_adversary_target_country   (independent of reports)
```

`vw_report`

Purpose: The main report view. It holds one record per RASD report, with the five single-valued classifications shown as labels. **Grain:** One row per report (`report_id` is unique). **Built from:** `fact_rasd_report` joined to `dim_rasd_classification`, `dim_rasd_evidence_type`, `dim_rasd_importance_level`, `dim_rasd_source`, `dim_rasd_threat_type`.

Column	Type	Description
`id`	INT	Fact surrogate key, assigned by ordering on `report_id`. It is renumbered on every rebuild, so don't store it outside the warehouse.
`report_id`	STRING	RASD report identifier (business key).
`title`	STRING	Report title.
`description`	STRING	Narrative description of the observation.
`actions`	STRING	Recommended or taken actions.
`analysis`	STRING	Analyst assessment.
`creation_date`	DATE	Date the report was created.
`publication_date`	DATE	Date the report was published.
`report_date`	DATE	Report (observation) date; the main date for trend analysis.
`updated_at`	TIMESTAMP	Last update timestamp in the source.
`classification`	STRING	Report classification, `'Unknown'` if missing.
`evidence_type`	STRING	Type of supporting evidence, `'Unknown'` if missing.
`importance`	STRING	Importance level (text label, not numeric), `'Unknown'` if missing.
`observation_source`	STRING	Where the observation came from, `'Unknown'` if missing.
`threat_type`	STRING	Threat category, `'Unknown'` if missing.
`country_group_key`	STRING(32)	Key to the report's country set (see `vw_report_country`).
`entity_group_key`	STRING(32)	Key to the report's entity set (see `vw_report_entity`).
`group_group_key`	STRING(32)	Key to the report's threat-group set (see `vw_report_group`).
`adversary_group_key`	STRING(32)	Key to the report's adversary set (see `vw_report_adversary`).

Usage notes

- Use this view for report counts, trends by date, and breakdowns by classification, importance, source or threat type.
- `importance` is text, so sort or bucket it by label. Don't `AVG` it.

`vw_report_country`

Purpose: Lists the countries each report concerns. **Grain:** One row per report × resolved country. **Built from:** `vw_report` → `bridge_country` (on `country_group_key`) → `dim_country`.

Column	Type	Description
`report_id`	STRING	RASD report identifier.
`title`	STRING	Report title.
`report_date`	DATE	Report date.
`classification`	STRING	Report classification.
`importance`	STRING	Importance level.
`threat_type`	STRING	Threat category.
`country_id`	BIGINT	Surrogate id in `dim_country`; `-1` = unknown or unmatched.
`country_name`	STRING	Country name; `'Unknown'` when `country_id = -1`.
`weight_factor`	DECIMAL(23,22)	`1 / number of countries on the report`.
`is_unknown_member`	BOOLEAN	`TRUE` if the report has no country, or a listed country didn't match `dim_country`.

Usage notes

- Every report has at least one row. Reports with no country get one `Unknown` row with `weight_factor = 1`.
- In the source `country` field, countries are **comma-separated**.

`vw_report_entity`

Purpose: Lists the organisations (entities) each report concerns, with their CTI register attributes. **Grain:** One row per report × resolved entity. **Built from:** `vw_report` → `bridge_entity` (on `entity_group_key`) → `dim_rasd_entity`.

Column	Type	Description
`report_id`	STRING	RASD report identifier.
`title`	STRING	Report title.
`report_date`	DATE	Report date.
`classification`	STRING	Report classification.
`importance`	STRING	Importance level.
`threat_type`	STRING	Threat category.
`entity_id`	BIGINT	Surrogate id in `dim_rasd_entity`; `-1` = unknown or unmatched.
`entity_name_en`	STRING	English name. For label-only entities this repeats the Arabic name.
`entity_name_ar`	STRING	Arabic name.
`cti_id`	STRING	Entity id in the CTI entity register; `NULL` for label-only entities.
`prm_id`	STRING	Entity id in PRM; `NULL` if not linked.
`sector`	STRING	Sector from the CTI register.
`category`	STRING	Category from the CTI register.
`entity_type`	STRING	Entity type from the CTI register.
`entity_country`	STRING	The entity's country from the CTI register. This is **not** the report's country.

Column	Type	Description
`is_cti_entity`	BOOLEAN	`TRUE` = registered CTI entity. `FALSE` = name taken only from report text (label-only).
`weight_factor`	DECIMAL(23,22)	`1 / number of entities on the report`.
`is_unknown_member`	BOOLEAN	`TRUE` if the report has no entity, or a listed entity couldn't be matched.

Usage notes

- Report entities are matched in priority order: `cti_id` → `prm_id` → Arabic name → English name (case-insensitive).
- Label-only entities (`is_cti_entity = FALSE`) come from Arabic names in reports that have no `entities_cti_id` and no register match. Register attributes (`sector`, `category`, ...) are `NULL` for them.
- Filter on `is_cti_entity = TRUE` to analyse registered organisations only.

`vw_report_group`

Purpose: Lists the threat groups named in each report. **Grain:** One row per report × resolved threat group. **Built from:** `vw_report` → `bridge_group` (on `group_group_key`) → `dim_rasd_group`.

Column	Type	Description
`report_id`	STRING	RASD report identifier.
`title`	STRING	Report title.
`report_date`	DATE	Report date.
`classification`	STRING	Report classification.
`importance`	STRING	Importance level.
`threat_type`	STRING	Threat category.
`group_id`	BIGINT	Surrogate id in `dim_rasd_group`; `-1` = unknown or unmatched.
`group_name`	STRING	Threat group name; `'Unknown'` when `group_id = -1`.
`weight_factor`	DECIMAL(23,22)	`1 / number of groups on the report`.
`is_unknown_member`	BOOLEAN	`TRUE` if the report names no group, or a group couldn't be matched.

Usage notes

- Groups come from the report's own `related_groups` field. For attribution based on CTI intelligence, use `vw_report_adversary` instead.

`vw_report_adversary`

Purpose: Lists the CTI adversaries linked to each report. **Grain:** One row per report × resolved adversary. **Built from:** `vw_report` → `bridge_adversary` (on `adversary_group_key`) → `dim_cti_adversary`.

Column	Type	Description
`report_id`	STRING	RASD report identifier.
`title`	STRING	Report title.
`report_date`	DATE	Report date.

Column	Type	Description
`classification`	STRING	Report classification.
`importance`	STRING	Importance level.
`threat_type`	STRING	Threat category.
`adversary_id`	BIGINT	**Surrogate** id in `dim_cti_adversary`; `-1` = none or unmatched.
`adversary_name`	STRING	Adversary name; `'Unknown'` when `adversary_id = -1`.
`weight_factor`	DECIMAL(23,22)	`1 / number of adversaries linked to the report`.
`is_unknown_member`	BOOLEAN	`TRUE` if no adversary references this report.

Usage notes

- The link comes from the CTI side: an adversary record lists the RASD report ids it relates to (`adversary.rasd_ids`). The report itself doesn't name adversaries.
- ⚠️ `adversary_id` here is the **surrogate BIGINT**. In `vw_adversary_target_country`, `adversary_id` is the **CTI natural key (STRING)**. To combine the two views, join on `adversary_name`, or go through `dim_cti_adversary` (`id` ↔ `adversary_id`).

`vw_adversary_target_country`

Purpose: Shows the countries each CTI adversary is known to target, straight from CTI adversary intelligence. No reports are involved. **Grain:** One row per adversary × target country. **Built from:** `dim_cti_adversary` → `bridge_adversary_country` (on `adversary_id` + `country_group_key`) → `dim_country`.

Column	Type	Description
`adversary_id`	STRING	CTI adversary identifier (**natural key**, not the surrogate).
`adversary_name`	STRING	Adversary name.
`country_group_key`	STRING(32)	Key to the adversary's set of target countries.
`country_id`	BIGINT	Surrogate id in `dim_country`.
`country_name`	STRING	Target country name.
`weight_factor`	DECIMAL(23,22)	`1 / number of target countries listed for the adversary`, counted **before** unmatched countries are removed.
`is_unknown_member`	BOOLEAN	Always `FALSE` in this view, because unknown members are filtered out.
`total_targeted_countries`	BIGINT	Number of matched target countries for the adversary (the rows shown).

Usage notes

- Unknown adversaries (`id = -1`) and unmatched countries are excluded.
- Weights are counted before unmatched countries are removed. So if a country didn't match, the adversary's `weight_factor` values add up to **less than 1**. Use `total_targeted_countries` if you need an exact denominator.
- Adversaries with no target country in CTI don't appear.

`vw_entity_coverage`

Purpose: Data-quality scorecard for how report entities match the CTI entity register. **Grain:** One row per metric (5 rows). **Built from:** `dim_rasd_entity`, `bridge_entity`, `vw_report_entity`.

Column	Type	Description
`metric`	STRING	Metric name (see below).
`value`	BIGINT	Metric value.

`metric`	Definition
`cti_entities`	Registered CTI entities in `dim_rasd_entity`.
`cti_entities_with_prm_id`	Registered CTI entities that also have a `prm_id`.
`label_only_entities`	Entities known only from report text (not in the CTI register).
`unresolved_report_members`	Distinct entity member sets (`entity_group_key`) that contain at least one unmatched entity.
`reports_with_no_known_entity`	Reports with **at least one** unknown or unmatched entity row. Despite the name, this also counts reports that have some matched entities.

Usage notes

- The metrics are counts, not ratios. Get a coverage rate by dividing, for example `cti_entities_with_prm_id / cti_entities`.